

Cryptography Reference

Introduction

Cryptography is the science of securing information and communication through mathematical techniques. It ensures confidentiality, integrity, authentication, and non-repudiation of data. Modern cybersecurity systems use cryptography to protect passwords, banking data, emails, and digital files.

Objectives of Cryptography

Confidentiality: Ensures only authorized users can access information.

Integrity: Protects data from unauthorized modification.

Authentication: Verifies user or system identity.

Non-Repudiation: Prevents denial of actions or communications.

Symmetric Key Cryptography

Symmetric encryption uses the same key for encryption and decryption. It is fast and efficient for encrypting large amounts of data. Examples include AES and DES.

Asymmetric Key Cryptography

Asymmetric encryption uses a public key for encryption and a private key for decryption. It is commonly used in secure communication and digital signatures. Examples include RSA and ECC.

AES Encryption

AES (Advanced Encryption Standard) is a widely used symmetric encryption algorithm supporting 128-bit, 192-bit, and 256-bit keys. AES-GCM mode provides encryption along with authentication and integrity protection.

PBKDF2 Key Derivation

PBKDF2 (Password-Based Key Derivation Function 2) converts passwords into strong cryptographic keys using repeated hashing, making brute-force attacks more difficult.

SHA-256 Hashing

SHA-256 is a cryptographic hashing algorithm that generates a fixed 256-bit hash value. It is used for password hashing, integrity verification, and blockchain security.

Applications of Cryptography

Cryptography is used in online banking, secure messaging, SSL/TLS websites, cloud storage, digital signatures, military systems, and file encryption software.

Cryptography in SecureShield Project

The SecureShield File Encryption and Decryption System uses AES-256 encryption, PBKDF2 key derivation, SHA-256 hashing, and AES-GCM mode to ensure secure storage and transmission of confidential data.

Conclusion

Cryptography is an essential part of cybersecurity. It protects sensitive data from unauthorized access and cyber threats using advanced encryption and hashing algorithms.
